

PIMCI — Prime-Indexed Multiplicity Cobordism Invariants

Abstract

We formalize a **prime-indexed** family of cobordism/diffeomorphism invariants intended to detect **smooth** structure on homotopy spheres and related manifolds. The core idea is to attach to a manifold a **Kirby-safe algebraic package** (an integral quadratic form together with a smooth refinement), reduce it to a **canonical normal form** under the equivalence generated by handle/Kirby moves (stable integral congruence + stabilization), and then extract **prime-power layers** via p-adic truncation projectors Π_{p^r} . A **multiplicity operator calculus** viewpoint is incorporated by representing cobordisms as **noncommuting operator words** acting on the carrier data, and by using a (soft) **resonance functional** as an adequacy metric during computation while invariance is ultimately enforced by canonicalization.

Status note. The canonicalization and prime-layer extraction are intended to be strict invariants once the normal form is fully specified. The resonance functional is optional and should be treated as a *computational stability/fit* score rather than the definition of equivalence.

1. Notation and setting

1.1 Bordism categories

Let $\mathbf{Bord}_n^{\text{spin}}$ denote the (symmetric monoidal) n-dimensional spin bordism category (variants: framed/oriented). Objects are closed n-manifolds with the chosen structure; morphisms are bordisms.

1.2 The target data category

Define a category **QuadRef** whose objects are pairs

$$K = (Q, \rho)$$

where:

- Q is an integral symmetric bilinear form (matrix) representing an intersection form or surgery kernel (depending on dimension/model), and
- ρ is a **smooth refinement datum** sufficient to reconstruct a smooth-sensitive invariant (e.g. Pontryagin class data, characteristic vectors, η /APS correction terms, or an Eells–Kuiper-type refinement).

Morphisms in **QuadRef** are generated by:

1. **Integral congruence** (basis change)

$$((Q, \rho) \mapsto (U^{\mathsf{T}} Q U, U^{\mathsf{T}} \rho U) \quad (U \in \mathrm{GL}(k, \mathbb{Z})),$$

1. **Stabilization / blow-up**

$$(Q, \rho) \mapsto (Q \oplus [\pm 1], \rho \oplus \rho_0),$$

with the corresponding inverse move when the summand is recognized as cancellable/trivial in the chosen theory.

We write $K \sim K'$ if they are related by the generated equivalence. This is the algebraic shadow of Kirby/handle moves.

2. Data extraction from a manifold

2.1 Input presentations

For a manifold M (especially a homotopy sphere Σ^n), choose one of:

- a framed link / Kirby diagram,
- a plumbing graph (e.g. for Milnor-type constructions),
- a handle decomposition.

From this presentation, produce a bounding cobordism W (when available) that carries the relevant structure (e.g. spin bounding for 7-spheres in the standard testbed).

Define

$$K(M) := (Q_W, \rho_W) \in \mathbf{QuadRef}.$$

Typical choices:

- Q_W : intersection form on a middle-dimensional homology group (e.g. $H_4(W; \mathbb{Z})$ for an 8-dimensional bounding W of a 7-manifold).
- ρ_W : refinement encoding smooth structure (examples include characteristic vectors, Pontryagin class evaluations, and/or data needed to compute an Eells–Kuiper-type invariant).

Design requirement (smooth sensitivity). ρ_W must not be purely homotopy/homology-level if the goal is to distinguish exotic smooth structures.

3. Canonicalization (the “gauge-fix”)

3.1 Canonical normal form

A **canonicalization map** is a function

$$\mathsf{NF}: \mathbf{QuadRef} \rightarrow \mathcal{K}$$

into a set $\mathcal{K}$ of canonical representatives such that:

- $\mathsf{NF}(K) = \mathsf{NF}(K')$ whenever $K \sim K'$, and
- NF is deterministic (independent of choices once the input K is fixed).

Informally, NF is a complete invariant of the Kirby-generated equivalence on (Q, ρ) within the chosen class of objects.

3.2 Implementable canonicalization scheme (algebraic)

A practical NF can be defined as a structured tuple of invariants and decompositions, for example:

- rank, signature, parity, discriminant of Q ,
- local p-adic Jordan decompositions of $Q \otimes \mathbb{Z}_p$ for primes in scope,
- a canonical representative of ρ modulo the automorphism group of Q (or relative to the p-adic decomposition).

Key constraint. Many spectral quantities (trace powers/eigenvalues) are *not* invariants of integral congruence. Canonicalization must happen *before* spectral-style postprocessing.

4. Prime-tier projectors

4.1 p-adic truncation projectors

For each prime power p^r , define a projector

$$\Pi_{\{p^r\}}: \mathcal{K} \rightarrow \mathcal{K}_{\{p^r\}}$$

interpreted as “retain information visible modulo p^r ” (or in $\mathbb{Z}_p$ up to depth r). Concretely, if $\mathsf{NF}(K)$ includes p-adic Jordan blocks and refinement components, Π_{p^r} keeps:

- the p-adic data of Q up to exponent r , and
- the refinement ρ reduced/normalized modulo p^r in a way compatible with the block decomposition.

4.2 Algebraic identities

The projectors satisfy:

- idempotence: $\Pi_{p^r}^2 = \Pi_{p^r}$,
 - filtration compatibility: $\Pi_{p^r} \circ \Pi_{p^{r+1}} = \Pi_{p^r}$,
 - coprime independence (conceptually): distinct primes act on independent localizations (formalized as a product of local data in NF).
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5. Multiplicity-weighted matrices (optional postprocessing)

Once $\text{NF}(K(M))$ is computed, one may define a **multiplicity-weighted matrix** representation

$$M(M) := D \cdot Q^{\text{NF}} \cdot D, \quad D := \text{diag}(p_1^{\mu_1}, \dots, p_k^{\mu_k})$$

where:

- Q^{NF} is the canonical form/matrix part extracted from $\text{NF}(K(M))$,
- $D = \text{diag}(p_1^{\mu_1}, \dots, p_k^{\mu_k})$ assigns primes p_i and exponents μ_i **deterministically from canonical block ordering**.

This preserves invariance because prime assignment occurs *after* canonicalization.

6. Definition of PIMCI (prime-indexed invariants)

6.1 The invariant family

Define the **Prime-Indexed Multiplicity Cobordism Invariants (PIMCI)** of M by

$$\mathfrak{I}_{p^r}(M) := F_{p^r} \left(\bigoplus_{\Pi \in \text{NF}(K(M))} \Pi \right),$$

where F_{p^r} is any deterministic functional on the p -tier canonical data.

Examples of allowed functionals (once in canonical form):

- invariants of the p -adic Jordan decomposition (block types, exponents, local signs),
- invariants derived from the refinement ρ modulo p^r ,
- spectral-style quantities computed from canonical matrices $M(M)$ (trace powers, characteristic polynomials) **only after** canonicalization.

6.2 Aggregation over primes

Optionally define an aggregated invariant

$$\mathfrak{I}(M) := \{\mathfrak{I}_{p^r}(M) \mid p \text{ prime}, r \leq r. \} (p)$$

In practice, only primes dividing the relevant classification torsion (e.g. those dividing $|\Theta_n|$ for homotopy spheres) contribute nontrivially in low dimensions.

6.3 Invariance statement (intended)

If $M \cong M'$ (diffeomorphic within the chosen structure) and the extraction $K(M)$ is compatible with handle/Kirby calculus, then

$$\mathfrak{I}_{p^r}(M) = \mathfrak{I}_{p^r}(M') \quad \text{for all } p^r.$$

Moreover, if M and M' are bordant and the theory is set up as a bordism functor, one expects $\mathfrak{I}_{p^r}$ to be constant on the corresponding bordism classes (depending on the chosen ρ).

7. Cobordisms as noncommuting operator words (MOC integration)

7.1 Carrier space

Represent the evolving presentation data as a carrier

$$\mathcal{M} = (X, H),$$

where H is a relational topology object (graph/hypergraph of generators and relations) and X holds state-like data (forms/refinements, block decompositions, etc.).

7.2 Prime-indexed operators

For each prime p and tier r , define noncommuting operators

$$\hat{O}^{(r)}_p: \mathcal{M} \rightarrow \mathcal{M}$$

that implement elementary transformations (slides, stabilizations, refinement updates) while tracking prime-tier structure.

A cobordism (or a sequence of handle moves) corresponds to a word

$$W \in \langle \hat{O}^{(r)}_p \rangle,$$

acting by $\mathcal{M} \mapsto W[\mathcal{M}]$. Noncommutativity reflects order-sensitivity of moves before reduction.

7.3 Strict invariance via reduction

Define the strict equivalence class by canonicalization:

$$[\mathcal{M}] := \text{NF}(K(M)) \sqcup \text{text}\{(\text{plus any additional canonical data}).\}$$

Then PIMCI is computed as a function of $[\mathcal{M}]$, ensuring move invariance.

8. Resonance functional (soft adequacy metric)

A resonance functional

$$R(H, \{\hat{O}\}; D) \in [0, 1]$$

may be used as a diagnostic during computation:

- H : relational topology (presentation hypergraph),
- $\{\hat{O}\}$: operator set/word constraints,
- D : dataset/target constraints (known invariants, consistency checks).

Recommended role:

- Guide search for canonical reductions (choose move sequences that increase R),
- Detect numerical instability or incomplete reduction,
- Provide a graded notion of “fit” before strict invariants are finalized.

Not recommended: defining equivalence of manifolds purely by maximizing R .

9. Specialization to 7-dimensional homotopy spheres (benchmark)

9.1 Known classification target

Let Θ_7 be the group of oriented diffeomorphism classes of smooth homotopy 7-spheres under connected sum. It is known that $\Theta_7 \cong \mathbb{Z}/28$. A classical smooth invariant (Eells–Kuiper μ) detects this structure in the parallelizable-bounding setting and provides a concrete benchmark for any proposed invariant family.

9.2 Prime-spectrum prediction

Since $28 = 4 \cdot 7$, low-tier nontrivial prime data should concentrate at $p = 2$ and $p = 7$. Thus a first-pass PIMCI for Θ_7 should reproduce (or strongly correlate with) $\mu \bmod 4$ and $\mu \bmod 7$ through $\mathfrak{I}_{2^r}$ and $\mathfrak{I}_{7^r}$ at small r .

10. Minimal viable experiment (MVE)

1. **Choose test objects:** explicit Milnor 7-spheres from plumbing or bundle presentations.

2. **Extract ***** $K = (Q, \rho)$ *****:** compute intersection matrix and refinement sufficient to obtain μ (or a μ proxy).
 3. **Canonicalize:** compute $\text{NF}(K)$.
 4. **Project:** compute Π_4 and Π_7 (or Π_{2^2}, Π_{7^1}).
 5. **Evaluate:** produce $\mathfrak{I}_4, \mathfrak{I}_7$ and compare to known μ residues.
 6. **Move-invariance check:** apply random unimodular congruences U and stabilization steps to (Q, ρ) ; verify invariants are unchanged after NF .
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11. Extensions and outlook

- **Beyond homotopy spheres:** apply to inertia groups / 2-connected 7-manifolds where generalized μ -invariants appear.
 - **Quantum gravity weighting:** define weights $w(M)$ from prime-spectrum invariants (e.g. root-of-unity phases) for toy “sum over smoothings.”
 - **String topology:** interpret Π_{p^r} as p-local shadows of loop product/operations, potentially revealing smooth refinements not visible in homotopy type alone.
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